PUSHFORWARD ALONG A COMPOSITION OF CONTINUOUS MAPS EQUALS THE COMPOSITION OF PUSHFORWARDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of pushforward: (g*f)(W) = F(g^{-1}W) for W open in Z. invoked at proof lines 2263-2266. Fixes the two evaluations being compared.

Lemma 0.2. *Let $f : X \to Y$ and $g : Y \to Z$ be continuous maps of topological spaces. The functors $(g \circ f)_*$ and $g_* \circ f_*$ are equal (on both presheaves and sheaves of sets).*

Proof. This is because $(g \circ f)_*\mathcal{F}(W) = \mathcal{F}((g \circ f)^{-1}W)$ and $(g_* \circ f_*)\mathcal{F}(W) = \mathcal{F}(f^{-1}g^{-1}W)$ and $(g \circ f)^{-1}W = f^{-1}g^{-1}W$. □

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